Neil Hwang
Linear Algebra
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Homework 2

Homework from Lecture on May 13, 2026

Find the Jordan canonical form of each of the following matrices.

Question (a).
Find the Jordan canonical form of

$$A = \begin{pmatrix} 1 & 1 \\ -1 & 3 \end{pmatrix}.$$

Solution.

Step 1: Characteristic polynomial.

$$\begin{aligned} \det(A - \lambda I) &= \det \begin{pmatrix} 1-\lambda & 1 \\ -1 & 3-\lambda \end{pmatrix} \\ &= (1-\lambda)(3-\lambda) - (1)(-1) \\ &= \lambda^2 - 4\lambda + 3 + 1 \\ &= \lambda^2 - 4\lambda + 4 \\ &= (\lambda - 2)^2. \end{aligned}$$

So $\lambda = 2$ is the only eigenvalue, with algebraic multiplicity 2.

Step 2: Geometric multiplicity.

$$A - 2I = \begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix}.$$

This matrix has rank 1, so $\nu_1(2) = \dim(\ker(A - 2I)) = 2 - 1 = 1$.

Step 3: Jordan form.
Since the algebraic multiplicity is 2 and the geometric multiplicity is 1, A is *not* diagonalizable. There must be a single Jordan block of size 2:

$$\boxed{J = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} = J_2(2).}$$

We can verify using the theorem from lecture: $2\nu_1 - \nu_2 - \nu_0 = 2(1) - 2 - 0 = 0$ blocks of order 1, and since the total size must be 2, there is one block of order 2.

Question (b).
Find the Jordan canonical form of

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}.$$

Solution.

Step 1: Characteristic polynomial.

$$\begin{aligned} \det(A - \lambda I) &= \det \begin{pmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{pmatrix} \\ &= (1-\lambda)(2-\lambda) - 6 \\ &= \lambda^2 - 3\lambda + 2 - 6 \\ &= \lambda^2 - 3\lambda - 4 \\ &= (\lambda - 4)(\lambda + 1). \end{aligned}$$

So the eigenvalues are $\lambda_1 = 4$ and $\lambda_2 = -1$, each with algebraic multiplicity 1.

Step 2: Jordan form.
Since both eigenvalues are distinct and have algebraic multiplicity 1, the geometric multiplicity of each is also 1. The matrix is diagonalizable, and its Jordan form consists of two 1×1 blocks:

$$\boxed{J = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix} = \operatorname{diag}\big(J_1(4),\, J_1(-1)\big).}$$

Verification: $\det J = (4)(-1) = -4 = \det A = (1)(2) - (2)(3) = -4$. $\operatorname{tr} J = 4 + (-1) = 3 = \operatorname{tr} A = 1 + 2 = 3$. ✓

Question (c).
Find the Jordan canonical form of

$$A = \begin{pmatrix} 11 & -4 & -5 \\ 21 & -8 & -11 \\ 3 & -1 & 0 \end{pmatrix}.$$

Solution.

Step 1: Characteristic polynomial.
We compute $\det(A - \lambda I)$. Expanding along the third row:

$$\det(A - \lambda I) = \det \begin{pmatrix} 11-\lambda & -4 & -5 \\ 21 & -8-\lambda & -11 \\ 3 & -1 & -\lambda \end{pmatrix}.$$

Expanding along the third row:

$$= 3\det\begin{pmatrix} -4 & -5 \\ -8-\lambda & -11 \end{pmatrix} - (-1)\det\begin{pmatrix} 11-\lambda & -5 \\ 21 & -11 \end{pmatrix} + (-\lambda)\det\begin{pmatrix} 11-\lambda & -4 \\ 21 & -8-\lambda \end{pmatrix}.$$

Computing each 2×2 determinant:

$$\det\begin{pmatrix} -4 & -5 \\ -8-\lambda & -11 \end{pmatrix} = 44 - 5(8+\lambda) = 4 - 5\lambda,$$
$$\det\begin{pmatrix} 11-\lambda & -5 \\ 21 & -11 \end{pmatrix} = -11(11-\lambda) + 105 = -16 + 11\lambda,$$
$$\det\begin{pmatrix} 11-\lambda & -4 \\ 21 & -8-\lambda \end{pmatrix} = (11-\lambda)(-8-\lambda) + 84 = \lambda^2 - 3\lambda - 4.$$

Combining:

$$\begin{aligned} \det(A - \lambda I) &= 3(4-5\lambda) + (-16+11\lambda) - \lambda(\lambda^2 - 3\lambda - 4) \\ &= 12 - 15\lambda - 16 + 11\lambda - \lambda^3 + 3\lambda^2 + 4\lambda \\ &= -\lambda^3 + 3\lambda^2 - 4. \end{aligned}$$

Testing $\lambda = -1$: $-(-1) + 3(1) - 4 = 1 + 3 - 4 = 0$. So $(\lambda + 1)$ is a factor.
Performing polynomial division:

$$-\lambda^3 + 3\lambda^2 - 4 = -(\lambda+1)(\lambda^2 - 4\lambda + 4) = -(\lambda+1)(\lambda-2)^2.$$

Eigenvalues: $\lambda_1 = -1$ (algebraic multiplicity 1), $\lambda_2 = 2$ (algebraic multiplicity 2).

Step 2: Geometric multiplicity of $\lambda = 2$.

$$A - 2I = \begin{pmatrix} 9 & -4 & -5 \\ 21 & -10 & -11 \\ 3 & -1 & -2 \end{pmatrix}.$$

Row reduce using $R_1 \leftarrow R_1 - 3R_3$ and $R_2 \leftarrow R_2 - 7R_3$:

$$\begin{pmatrix} 0 & -1 & 1 \\ 0 & -3 & 3 \\ 3 & -1 & -2 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - 3R_1} \begin{pmatrix} 0 & -1 & 1 \\ 0 & 0 & 0 \\ 3 & -1 & -2 \end{pmatrix}.$$

Rank $= 2$, so $\nu_1(2) = 3 - 2 = 1$.

Step 3: Jordan form.
Since $\lambda = -1$ has algebraic multiplicity 1, it contributes one $J_1(-1)$ block.
Since $\lambda = 2$ has algebraic multiplicity 2 and geometric multiplicity 1, it contributes one $J_2(2)$ block (not diagonalizable for this eigenvalue).

$$\boxed{J = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix} = \operatorname{diag}\big(J_1(-1),\, J_2(2)\big).}$$

Verification: $\det J = (-1)(2)(2) = -4$. $\det A = 11(-8 \cdot 0 - (-11)(-1)) - (-4)(21 \cdot 0 - (-11)(3)) + (-5)(21(-1) - (-8)(3)) = 11(-11) + 4(33) - 5(-21 + 24) = -121 + 132 - 15 = -4$. ✓
$\operatorname{tr} J = -1 + 2 + 2 = 3 = 11 + (-8) + 0 = \operatorname{tr} A$. ✓

Question (d).
Find the Jordan canonical form of

$$A = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 1 & -1 & 3 \end{pmatrix}.$$

Solution.

Step 1: Characteristic polynomial.

$$A - \lambda I = \begin{pmatrix} 2-\lambda & 1 & 0 & 0 \\ 0 & 2-\lambda & 1 & 0 \\ 0 & 0 & 3-\lambda & 0 \\ 0 & 1 & -1 & 3-\lambda \end{pmatrix}.$$

Expanding along the first column (only one nonzero entry in column 1):

$$\det(A - \lambda I) = (2-\lambda)\det \begin{pmatrix} 2-\lambda & 1 & 0 \\ 0 & 3-\lambda & 0 \\ 1 & -1 & 3-\lambda \end{pmatrix}.$$

Expanding the 3×3 determinant along the third column:

$$\det \begin{pmatrix} 2-\lambda & 1 & 0 \\ 0 & 3-\lambda & 0 \\ 1 & -1 & 3-\lambda \end{pmatrix} = (3-\lambda)\det \begin{pmatrix} 2-\lambda & 1 \\ 0 & 3-\lambda \end{pmatrix} = (3-\lambda)(2-\lambda)(3-\lambda).$$

Therefore:

$$\det(A - \lambda I) = (2-\lambda)^2(3-\lambda)^2.$$

Eigenvalues: $\lambda_1 = 2$ (algebraic multiplicity 2), $\lambda_2 = 3$ (algebraic multiplicity 2).

Step 2: Geometric multiplicity of $\lambda = 2$.

$$A - 2I = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & -1 & 1 \end{pmatrix}.$$

Row reduce: $R_3 \leftarrow R_3 - R_2$, then $R_4 \leftarrow R_4 - R_1$, then $R_4 \leftarrow R_4 + R_2$:

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Rank $= 3$, so $\nu_1(2) = 4 - 3 = 1$.
Since algebraic multiplicity $= 2$ and geometric multiplicity $= 1$, we get one $J_2(2)$ block.

Step 3: Geometric multiplicity of $\lambda = 3$.

$$A - 3I = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{pmatrix}.$$

Row reduce: $R_4 \leftarrow R_4 + R_2$:

$$\begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Rank $= 2$, so $\nu_1(3) = 4 - 2 = 2$.

Since algebraic multiplicity $= 2$ and geometric multiplicity $= 2$, $\lambda = 3$ is diagonalizable and contributes two $J_1(3)$ blocks.

Step 4: Jordan form.

$$\boxed{J = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} = \operatorname{diag}\bigl(J_2(2),\ J_1(3),\ J_1(3)\bigr).}$$

Verification:

- $\det J = 2 \cdot 2 \cdot 3 \cdot 3 = 36$. $\det A = 2 \cdot 2 \cdot 3 \cdot 3 = 36$ (expanding along column 1, then noting the upper triangular-like structure). ✓
- $\operatorname{tr} J = 2 + 2 + 3 + 3 = 10 = 2 + 2 + 3 + 3 = \operatorname{tr} A$. ✓
- Characteristic polynomial: $(2 - \lambda)^2 (3 - \lambda)^2$. ✓